

Supplementary Information for Phylogenomics of a genus of ‘Great Speciators’ reveals rampant incomplete lineage sorting, gene flow, and mitochondrial discordance in island systems

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The following Supplementary Information includes:

Supplementary Methods and Discussion

Supplementary figures 1–27

Supplementary Methods

Sampling and laboratory methods

For tissue-sourced samples, we used the Qiagen DNeasy kit to extract genomic DNA. For toepad-sourced samples, we performed phenol-chloroform extractions with the addition of Dithiothreitol (DTT) and gel phase-lock tubes (Murphy and Hellwig 1996) to increase yields for toepad samples, following best practices (Billerman and Walsh 2019; Tsai et al. 2019). We estimated fragment length with gel electrophoresis for tissue-sourced samples and sheared 250 ng of genomic DNA, aiming for 500 base-pair (bp) fragments, with a Covaris M220 focused-ultrasonicator (50 W peak incident power, 10% duty factor, and 200 cycles per burst for 55–75 s, dependent on sample quality). Toepad-sourced extractions are known to have much smaller average fragment sizes (Tsai et al. 2019) and therefore do not need to be sheared further if older than 30 years. Instead, we performed a Uracil-damage treatment (Rohland et al. 2015) with USER enzyme (New England Biolabs) to remove deaminated cytosines from all toepad-sourced extractions prior to library preparation. This method reduces natural DNA damage patterns and has been used for other projects reliant on whole-genome resequencing of toepad-sourced samples (Irestedt et al. 2022). We quantified DNA extractions with a Qubit 3.0 Fluorometer (ThermoFisher Scientific) for all samples prior to library preparation.

We prepared Illumina libraries with half the volume per sample using the Kapa Biosystems Hyper Prep Kit and the iTruStub dual-indexing system (baddna.org). For toepad-sourced samples, we did not sonicate prior to library prep, used Eppendorf Lo-Bind tubes for all non-thermocycler reactions, doubled ligation time (30 min), and increased concentrations of all AMPure (Beckman Coulter) bead cleanups to 3X sample volume. For all samples, we used the Qiagen GeneRead Size Selection Kit to remove fragments <150 bp prior to sequencing and a Bioanalyzer to visualize fragment sizes. We pooled toepad- and tissue-sourced libraries separately (10–15 toepad libraries per lane and 15–16 tissue libraries per lane) and sequenced them across 12 S4 PE-150 lanes of an Illumina NovaSeq 6000 at the Oklahoma Medical Research Foundation.

Removal of “dirty ends” of UCE loci

During preliminary phylogenetic exploration, we determined that many of the toepad-sourced samples had exceptionally long branches (see below for phylogenetic analyses and Table S1 for

these samples). Biologically unrealistic long branches such as these can contribute to long-branch attraction and therefore incorrect phylogenetic relationships (Felsenstein 1978); this issue has been shown as a byproduct of poor alignment trimming contributing to “dirty ends” of UCE loci sourced from historical samples, including those sourced from avian toepads (McCormack et al. 2015; Smith et al. 2020; Salter et al. 2022). We removed these artifactual “dirty ends” following the bioinformatic pipeline by Smith et al. (2020). To do so, we chose clade-specific reference samples to which we aligned trimmomatic cleaned reads; to identify these reference samples, we expanded fasta files (with “`phyluce_assembly_explode_get_fastas_file`”) and chose a closely related, tissue-sourced sample with the highest number of UCE loci based on initial maximum likelihood analyses (Table S2). We used `bwa` and `SAMtools` to index the UCE contigs of reference samples and align cleaned reads; we removed low-quality data in the flanking regions by dropping sites with $<5X$ coverage and quality scores of <20 . We manually incorporated these cleaned samples back into the `phyluce` pipeline by adding the nucleotide data into a combined unaligned fasta file (with all other non-cleaned samples) and again aligned and trimmed loci with the same settings as above. Our final UCE dataset was a 90% complete matrix comprising multiple individuals for some taxa that had at least 105 of 117 samples present at each UCE locus (Table S3).

Filtering of UCE and BUSCO loci for ASTRAL

We used the summary-statistic method ASTRAL-III v.1.15.2.4 (Zhang et al. 2018; Rabiee et al. 2019) to estimate the species tree for UCE and BUSCO loci. This summary-statistic MSC method is sensitive to fragmentary data and depends on high quality gene trees (Mirarab 2023). We considered quality of gene trees at a per-sample and per-locus basis and used different filtering methods to address each issue for both molecular markers. For example, a toepad-sourced sample could be present in thousands of loci but have a high degree of missing data (either coded as completely ambiguous or as a gap) for some but not all loci. Prior to estimation of gene trees, we filtered samples from individual alignments if they comprised significant missing data. For the UCE dataset, we dropped samples the 90% complete matrix if they comprised more than 25% of completely ambiguous nucleotides (N; `drop_fasta_N.py`). Then we dropped samples if they comprised more than 70% of gaps (-). From these filtered sets of alignments, we inferred gene trees with `IQtree2`. Because every sample was present at each

BUSCO locus, we increased the thresholds to filter samples and dropped samples from individual alignments if the sample comprised more than 50% of completely ambiguous nucleotides (X's) using custom python scripts (drop_fasta_X.py available on dryad). For both UCE and BUSCO locus specific datasets, we performed model selection and assessed gene tree support with 1,000 ultrafast bootstraps in IQtree2.

Gene tree quality could also be dependent on the informativeness of the locus, as some loci will have fewer phylogenetically informative sites than others. Therefore, alignments comprising fewer parsimony informative sites might be lower quality than those with many informative sites. Therefore, we filtered the IQtree-inferred gene trees based on the number of parsimony informative (PI) sites. For UCEs, we calculated the number of PI sites with `phyluce_align_get_informative_sites` and skimmed the number of PI sites for the BUSCO dataset from the log file of the partitioned IQtree2 analysis. For our UCE dataset, a threshold of >50 PI sites (1,583 gene trees) inferred a clear artifactual pattern in which the toepad samples clustered at the base of clades to the exclusion of tissue-sourced genomes (Fig. S2), a known issue for ASTRAL that is caused by poor quality and/or uninformative gene trees (Mirarab 2023) and the inclusion of DNA from degraded sources (Smith et al. 2020). Therefore, we further filtered UCE gene trees based on loci with at least 100 PI sites, which yielded 125 highly informative gene trees. For our BUSCO dataset, a PI threshold of 20 PI amino acid sites (resulting in 4,177 gene trees) reflected a similar artefactual clustering pattern of toepad-sourced samples (Fig. S4). This artefactual pattern disappeared with a PI threshold of 100 PI AA sites (470 gene trees; Fig. S15).

Supplementary Discussion

Gene flow within Todiramphus kingfishers

Sacred Kingfisher (*T. sanctus*) is one of two truly migratory species of *Todiramphus*, with a breeding range across Australia and a non-breeding range spanning Sumatra to the Solomon Islands (Woodall 2001). Although its widely considered just an Australian breeder, a few reports suggest that this species may breed outside of Australia (Dutson 2011). Given our finding of gene flow between *T. sanctus* and *T. sacer*—a non-migratory species endemic to Pacific islands in Melanesia to Samoa—evaluating these reports provides context for our genomic results.

Prior to this study, there were two pieces of evidence that the migratory *T. sanctus* breeds outside of Australia. The first is a very brief record (French 1957) that only lists a clutch size of three eggs for *T. sanctus* on the Olu Malau Islands (also known as the Three Sisters Islands, small islands off the northeastern coast of Makira, Solomon Islands). However, this record lists three breeding *Todiramphus* (then treated under “*Halcyon*”): *T. saurophagus*, *T. sanctus*, and *T. chloris*. At the time, “*Halcyon chloris*” could refer to two taxa that occur on the Three Sisters today but are now recognized as distinct species: *T. tristrami alberti* (widespread in the Solomon Islands) or *T. sacer sororum* (endemic to the Three Sisters; Woodall 2001; Gill et al. 2024; Fig. S1). These two taxa are morphologically distinct. *T. sacer sororum* resembles other *T. sacer*, with white underparts, a white collar, and a sexually dichromatic superciliary (“eyebrow”) stripe that extends over the eye and around the back of the head (white in females, rufous in males; see Figs. S25–27). In contrast, *T. tristrami alberti* has dark buff underparts and no eyebrow stripe, a phenotype much more similar to *T. sanctus*, which also has buff underparts, no eyebrow, but varying degrees of scalloping on the breast (Fig. S27). Given this similarity, *T. t. alberti* could easily have been mistaken for *T. sanctus* in the field. French (1957) did not describe how he determined species identity, leaving the record ambiguous. Thus, the clutch reported from the Olu Malau Islands is more plausibly explained as a simple field misidentification of two difficult sympatric species than as the first and only confirmed case of *T. sanctus* breeding outside Australia.

The second source of evidence stems from the only other phylogeographic study of the genus in which two individuals collected outside of the range of *T. sanctus* (in Temotu, Solomon Islands and Vanuatu) were inferred to be nested within the *T. sanctus* clade (Fig. 2 from Andersen et al. 2015b). In that study, two birds from Nendo Island were inferred with mtDNA to be in separate clades, with one in *T. sanctus* (KU tissue #19403) and the other with *T. sacer* (KU 19404). Because of its placement in the *T. sanctus* clade, Andersen et al. (2015) considered KU 19403 a misidentified *T. sanctus* rather than *T. sacer ornatus* with *T. sanctus* mtDNA (hence it’s identification as *T. sanctus* in their Fig. 2). However, comparisons of the specimen in question confirm it is not a case of misidentification because plumage characters match other *T. sacer ornatus* (a male with a rufous eyebrow and unspotted rufous breast matching other *T. sacer ornatus* collected during the Whitney South Seas expedition, Fig. S25). We sequenced the “*T. sanctus*” from Nendo in this study (KU 19403; Table S1) and found that nuclear data supported

its position within *T. sacer* but mitochondrial data placed it again in *T. sanctus* (Fig. 4). Clearly, the mitochondrial capture of *T. sanctus* mtDNA for *T. sacer ornatus* is not complete: both Andersen et al. (2015) and this study showed that other *T. sacer ornatus* are correctly grouped with other *T. sacer* (both KU 19404 in Andersen et al. 2015 and the AMNH toepad 217314 collected in the 1920's were inferred to group with other *T. sacer*). Andersen et al. (2015) also found that a *Todiramphus* collected from Espiritu Santo, Vanuatu, grouped within the *T. sanctus* clade (tissue number of B45812 at the Louisiana State University Museum of Natural Science). We did not sample this individual, but comparisons of the specimen (skin number 42929 housed at the University of Florida Museum of Natural History) show that its plumage matches that of other *T. sacer santoensis* (endemic to Vanuatu) and not of *T. sanctus* (Fig. S26). We predict that the nuclear data of this individual would similarly be placed with other *T. sacer* (like *T. sacer ornatus*, described above) and not *T. sanctus*. Taken together, the genetic evidence also points to a lack of current breeding records of *T. sanctus* outside of Australia.

Other sources of topological discordance

Another step taken to decrease the methodological influence of data source dealt with the extraction of BUSCO loci from toepad-sourced genomes. We found that we recovered poor quality BUSCO data when we extracted these markers from toepad-sourced genomes. This pattern was easily distinguishable because they clustered together in phylogenetic analyses in a non-biological pattern, similar to the situation noted above (Fig. S3). When we instead extracted BUSCO loci from coordinates from loci from our reference genome, this artifactual pattern disappeared and toepad-sourced genomes no longer clustered together (Figs. S15–16). Though more often used to determine the general completeness of a genome (Simão et al. 2015; Seppey et al. 2019; Feng et al. 2021), BUSCO loci are less frequently used as molecular markers in avian phylogenetics (but see Alaei Kakhki et al. 2023; Musher et al. 2024). Relatively few studies, to our knowledge, have extracted BUSCO loci from genomes sourced from degraded DNA for use in phylogenetic inferences (Musher et al. 2024). Considering the exponential increase in phylogenomic studies that include degraded DNA sourced from museum samples (“museumomics”; Raxworthy and Smith 2021; Ernst et al. 2022), future studies that aim to incorporate BUSCOs from degraded sources may want to explore alternate methods of extraction like those in this study.

Taxonomic recommendations

We found that *T. enigma*, an endemic of the Talaud Islands, Indonesia, is closely related to nominate *T. chloris*, often to the exclusion of the rest of the *T. chloris* taxa (Figs. 1, S8–10, S15–17). Originally described as a distinct species (Hartert 1904), *T. enigma* was long considered a subspecies of *T. chloris* until it was split based on phenotypic and life history differences (Riley et al. 1998). Since its elevation to species status, there have been relatively few studies on its ecology and no further information on the breeding biology of *T. enigma* or *T. chloris* on Talaud (Riley 2003; Kelly et al. 2017). *Todiramphus enigma* was not sampled by Andersen et al. (Andersen et al. 2015) and higher-level analyses that included *T. enigma* lacked sufficient sampling to place this taxon unequivocally (Andersen et al. 2018; McCullough et al. 2019). Our results suggest that *T. enigma* is best treated as part of *T. chloris*; however, a more nuanced explanation may involve undetected gene flow between these sympatric species or due to a toepad artifact, since our sample of *T. enigma* was sourced from a specimen from 1886 with 6.8x average coverage. Considering that *T. enigma* is a near-threatened species endemic to three small islands (~1,250 km² total land area), it is susceptible to threats from ongoing deforestation. More focused research is needed before lumping *T. enigma* with *T. chloris*.

We consistently found that two *T. sacer* taxa were embedded within the *T. tristrami* clade (Figs. 1, S8–17). *Todiramphus sacer pavuvu* and *mala* are some of the most northern representatives of *T. sacer*, endemic to Pavuvu (also referred to as the Russell Islands) and Malaita, respectively, in the Solomon Islands. *Todiramphus tristrami* and *T. sacer* were, among other *Todiramphus* species (*T. sordidus*, *T. saurophagus*, *T. colonus*, *T. albicilla*), part of the *T. chloris* species complex when it was thought to comprise 49 subspecies prior to Andersen et al. (2015). In that study, neither of these taxa were sampled and their species identification was likely based on geographic range and plumage characters. Here, *T. sacer pavuvu* and *mala* were sourced from toepads that were inferred to have a close association with *T. tristrami alberti*, which is found across the main Solomons Archipelago. Morphologically, these two taxa are quite similar to *T. tristrami alberti*, with their original species descriptions only highlighting minute differences from *alberti* (Mayr 1935). Considering their non-monophyly with other *T. sacer*, we recommend treating *pavuvu* and *mala* within *T. tristrami*. Whether or not *pavuvu* and *mala* are best treated as valid subspecies remains an outstanding question. To our knowledge,

mala is the only *T. tristrami* taxon on Malaita, an island that has several other endemic bird taxa, so it seems reasonable to treat *mala* as different from *alberti*. The question of *pavuvu* is a little more uncertain because we do not know if true *alberti* occurs on the Pavuvu Islands. It is unlikely that *alberti* and *pavuvu* co-occur on these tiny islands, but until this point can be resolved, we recommend maintaining *pavuvu* as a subspecies within *T. tristrami*.

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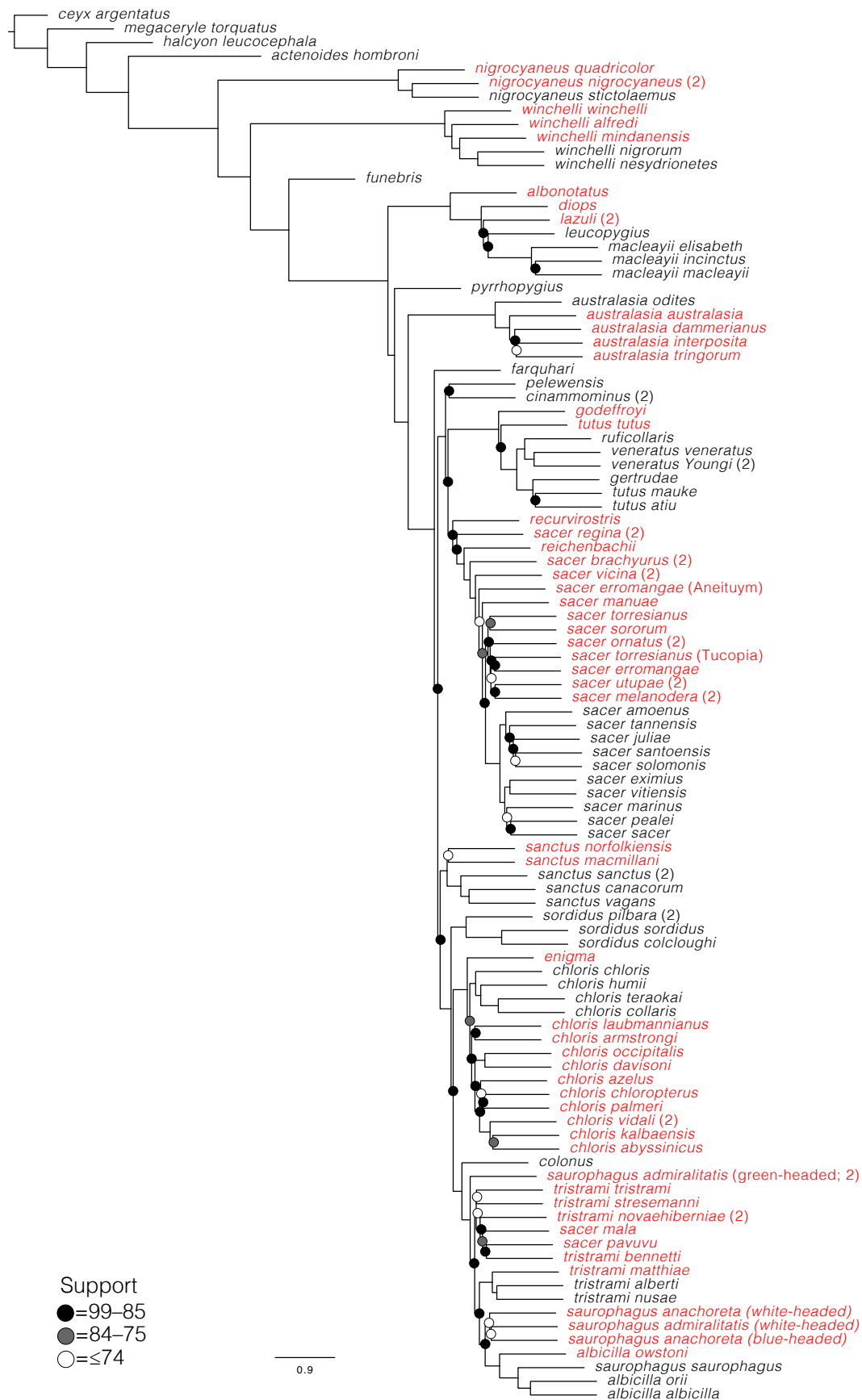


Figure S2: ASTRAL phylogeny of UCE 90% complete matrix with a threshold of >50 parsimony informative sites (1583 gene trees). Red names are toepad-sourced samples and the following number in parenthesis indicates how many samples are represented in that tip that were separate in gene trees (lumped by taxon).



Figure S3: IQtree phylogeny of 8,012 BUSCO loci when extracted from genomes using the BUSCO program. Toepad-sourced samples are red with an asterisk (*). Note the artificially long branches and artefactual clustering toepad samples.

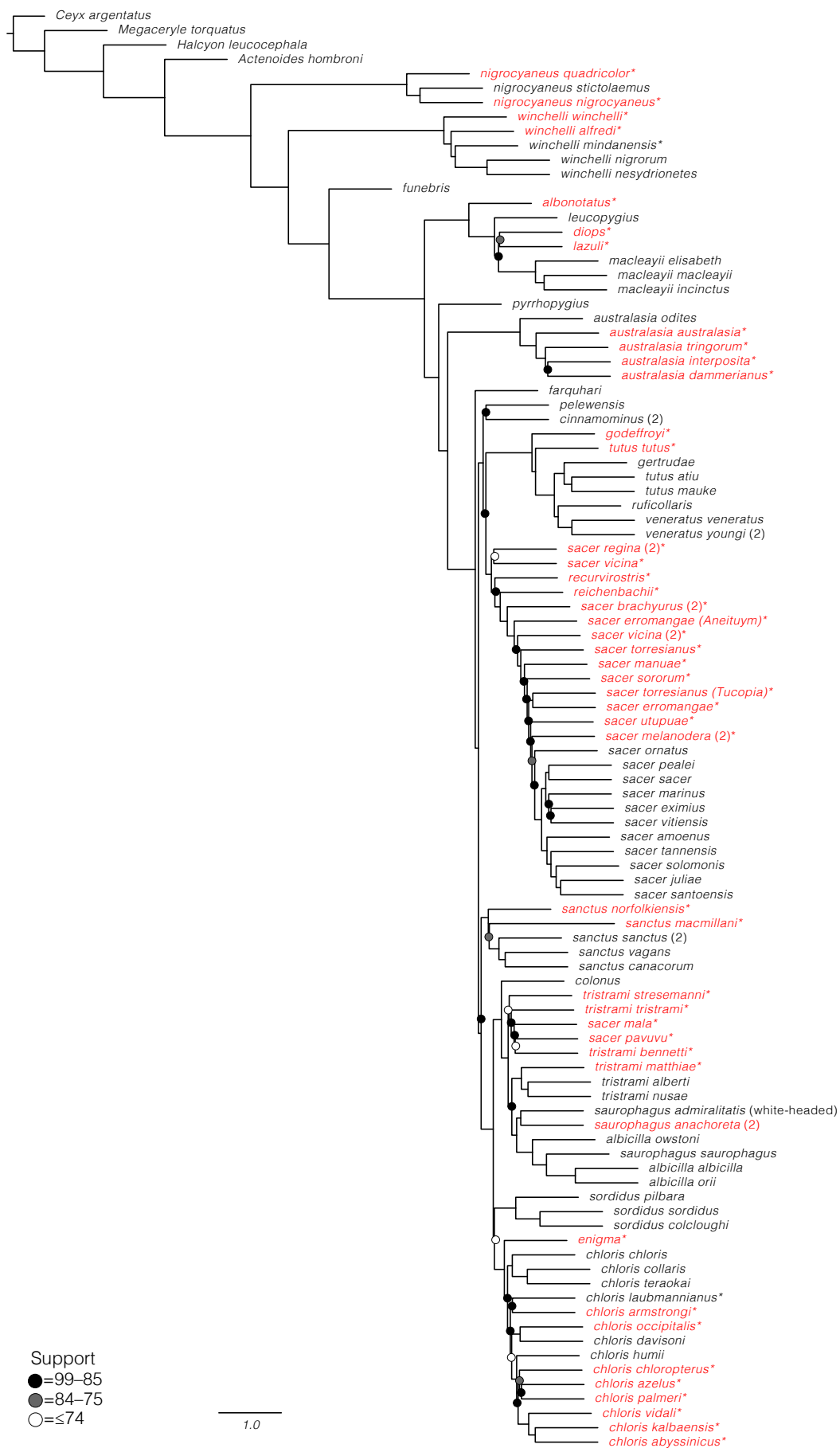


Figure S4: ASTRAL phylogeny of BUSCO loci with a threshold of >20 parsimony informative sites (4,177 gene trees). Red names are toepad-sourced samples and the following number in parenthesis indicates how many samples are represented in that tip that were separate in gene trees (lumped by taxon).

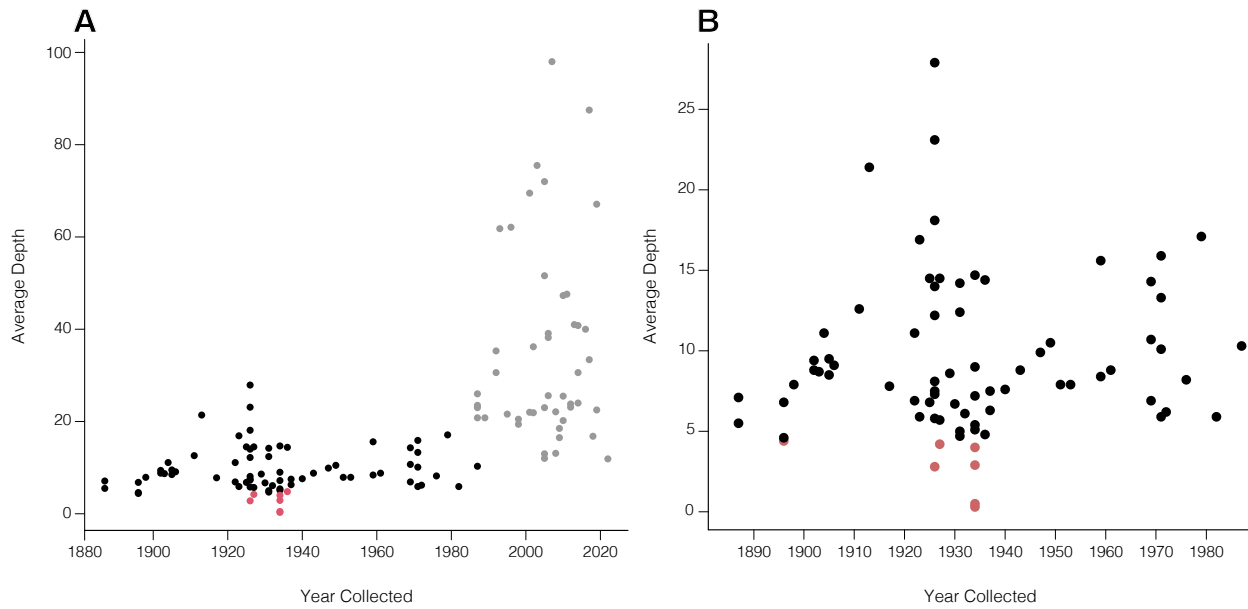


Fig S5. Average depth of all samples (A) and just toepad-sourced samples (B). Dot color indicate source of sample; gray dots are tissue-sourced, black dots are toepad-sourced, and red dots are toepad-sourced samples that were sequenced and successfully assembled but were not included in downstream analyses.

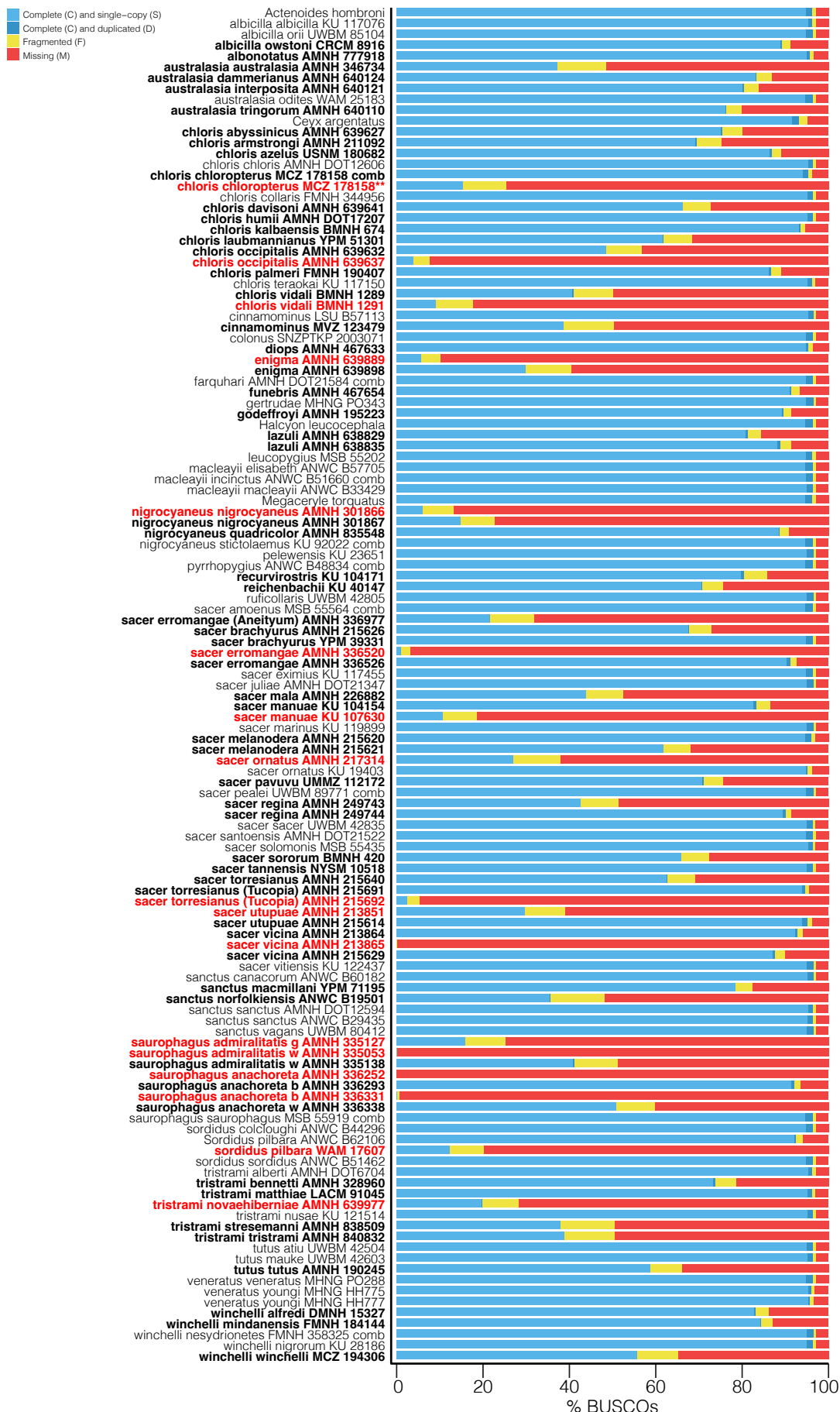


Figure S6: BUSCO completeness graph of all samples in this study. Light blue indicates complete and single-copy loci. Dark blue is complete and duplicated loci. Yellow is fragmented loci. Red is missing loci. Sample names that are bolded are toepad-sourced samples and red are samples that were not included in the BUSCO dataset.

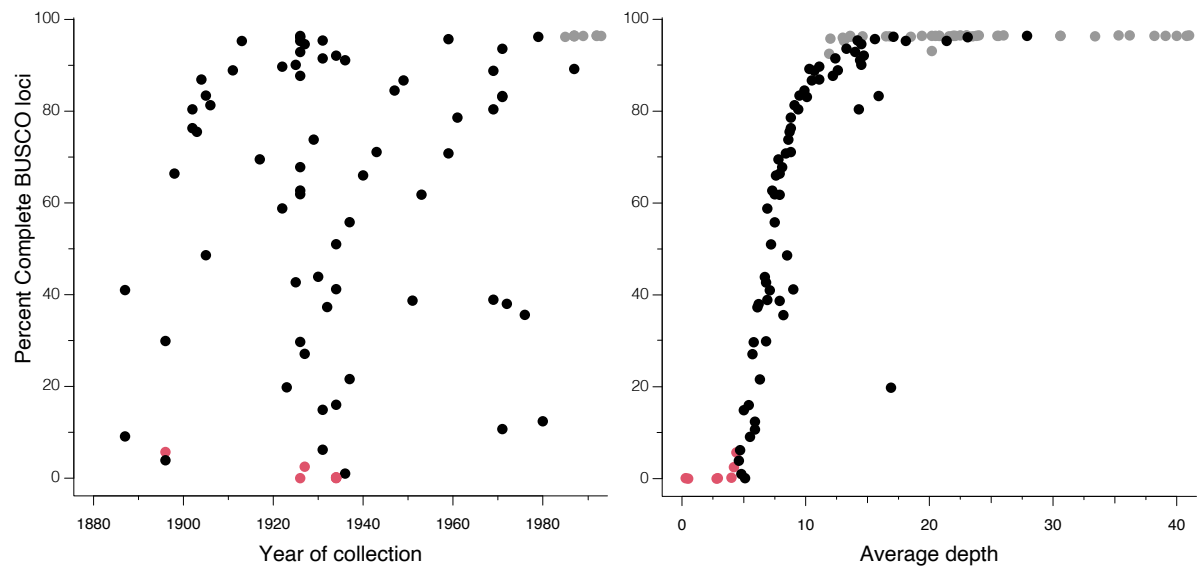


Figure S7: Average depth, not collection year, was more important for percent complete single-copy BUSCO loci for toepad-sourced genomes (out of 8,338 single-copy BUSCO loci). Dot color represents tissue-sourced (grey), toepad-sourced (red), and toepad-sourced samples that were not included in any subsequent analyses (red). We show only a subset of years and depths relevant to toepad-sourced genomes.

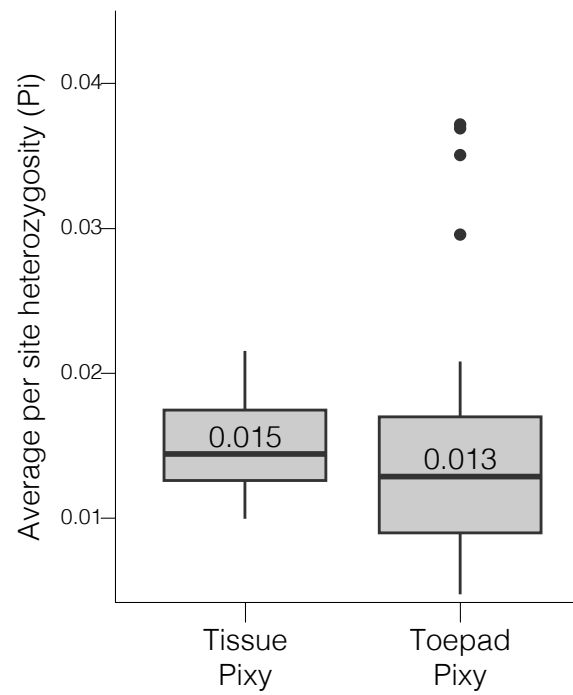


Figure S8: Comparison of average per site heterozygosity (π) estimated from toepad and tissue-sourced genomes with Pixy.

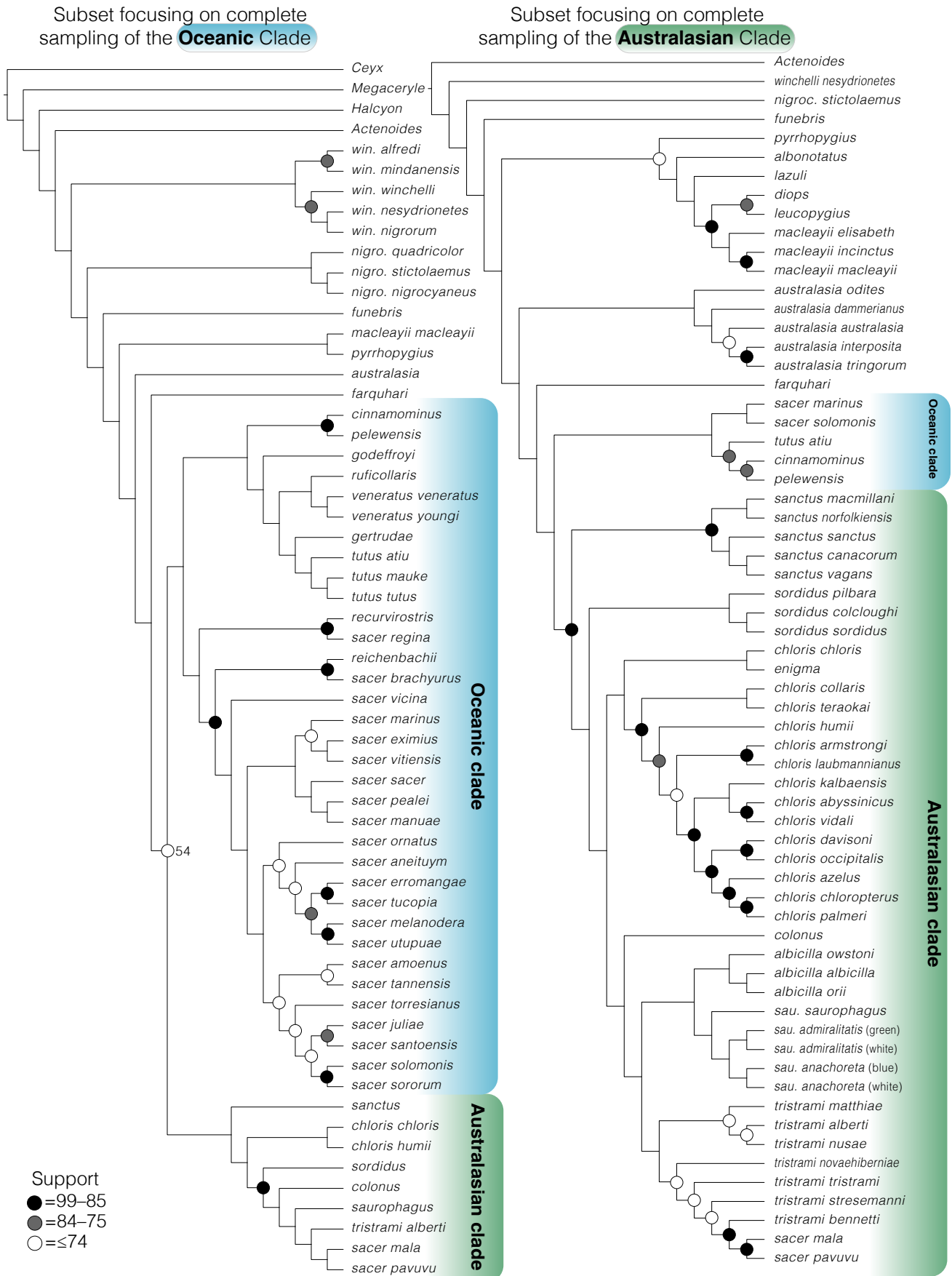


Figure S9: Subsets of the species tree, produced in SVDQuartets, from the 90% complete UCE matrix (4,942 loci, 4,531,264 bp). Circles reflect node support values that are below 100 BS: black circles indicate 99–85 BS, gray circles indicate 84–75 support, and white circles indicate 74 and lower support.

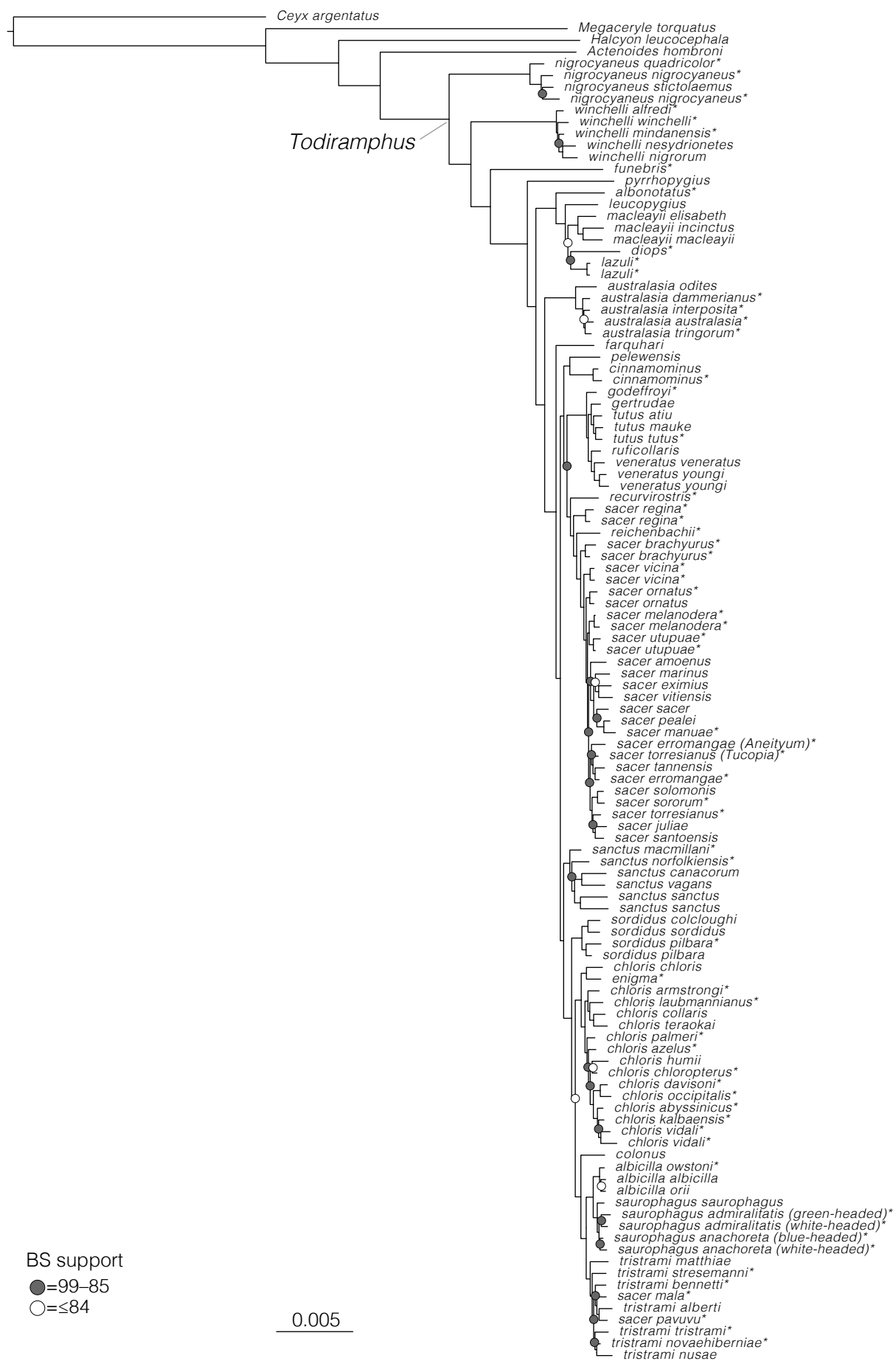


Figure S10: Partitioned maximum likelihood phylogeny, produced in IQtree, from the 90% complete UCE matrix (4,942 loci, 4,531,264 bp). Circles reflect node support values that are below 100 BS: gray circles indicate 99–85 support and white circles indicate 84 or lower support. Samples with asterisks (*) indicate toepad-sourced samples.

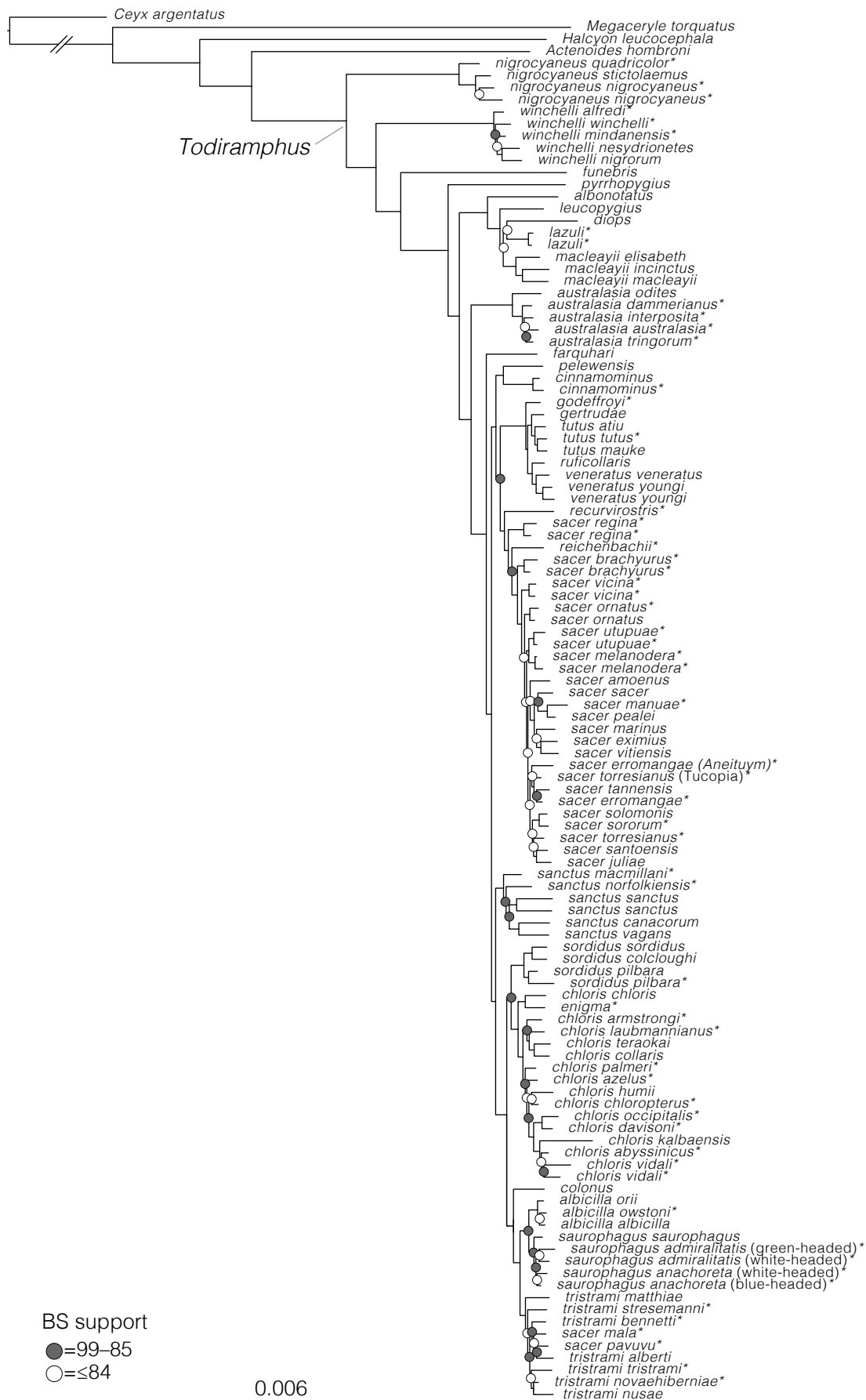


Figure S11: Maximum likelihood phylogeny, produced in RAxML-ng, from the 90% complete UCE matrix (4,942 loci, 4,531,264 bp). Circles reflect node support values that are below 100 BS: gray circles indicate 99–85 support and white circles indicate 84 or lower support.

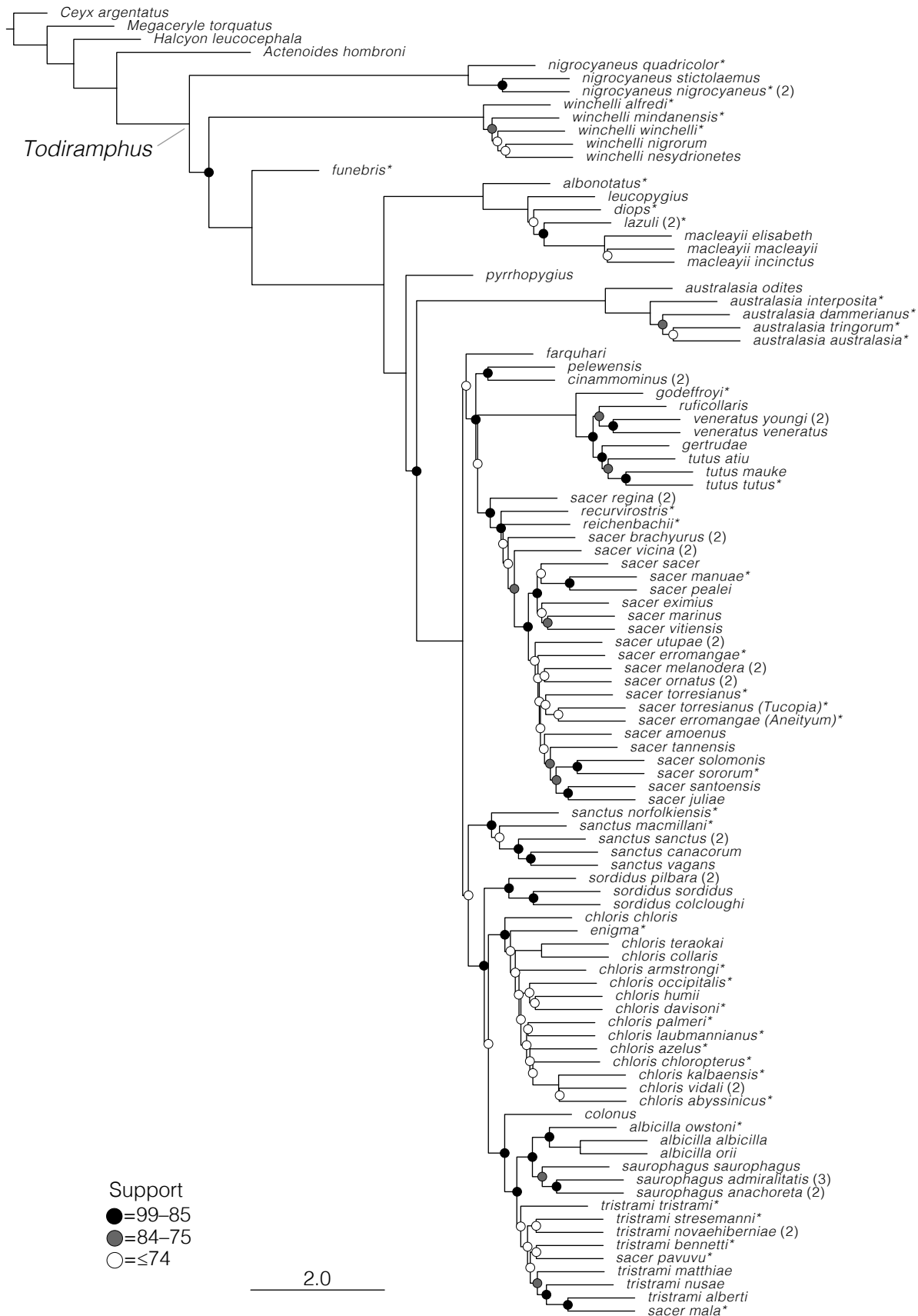


Figure S12: Species tree of *Todiramphus* kingfishers, produced in ASTRAL-III, based off of loci from the 90% complete that had at least 100 parsimony informative sites (125 loci). Circles reflect node support values that are below 100 BS: black circles indicate 99–85 BS, gray circles indicate 84–75 support, and white circles indicate 74 and lower support. Samples with asterisks (*) indicate toepad-sourced samples. Tips with numbers in parentheses indicate multiple samples that were distinct in gene trees but were lumped as a single tip for ASTRAL.

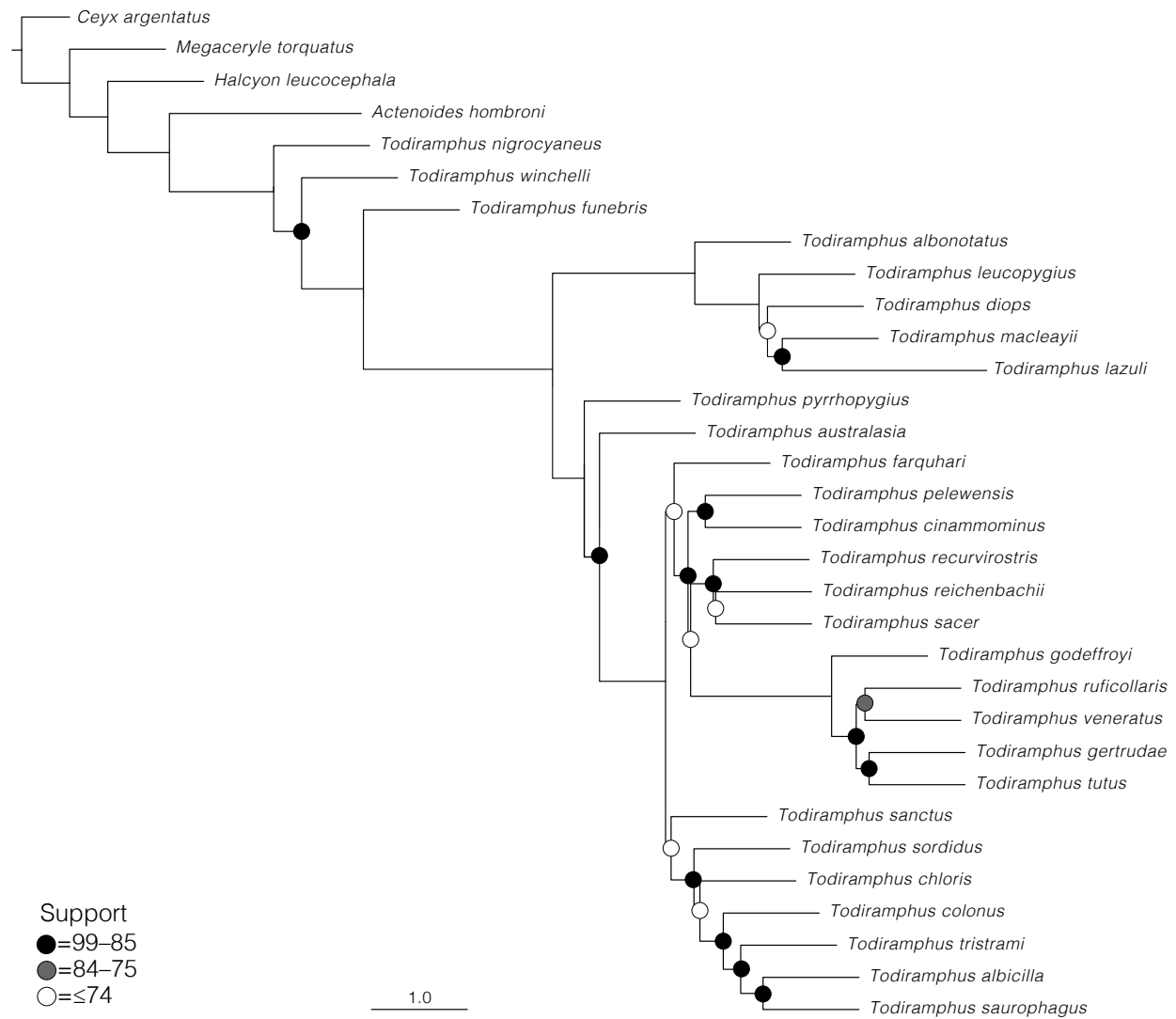


Figure S13: Species tree of *Todiramphus* kingfishers, produced in ASTRAL-III, based off of loci from the 90% complete that had at least 100 parsimony informative sites (125 loci). Circles reflect node support values that are below 100 BS: black circles indicate 99–85 BS, gray circles indicate 84–75 support, and white circles indicate 74 and lower support. See Fig S12 for all subspecific diversity.



Figure S14: SNP-based species tree, produced in SVDQuartets, from the 100% complete SNP data-set (85,156 SNPs) for 119 tips. Circles reflect node support values that are below 100 BS: black circles indicate 99–85 BS, gray circles indicate 84–75 support, and white circles indicate 74 and lower support. Samples with asterisks (*) indicate toepad-sourced samples.

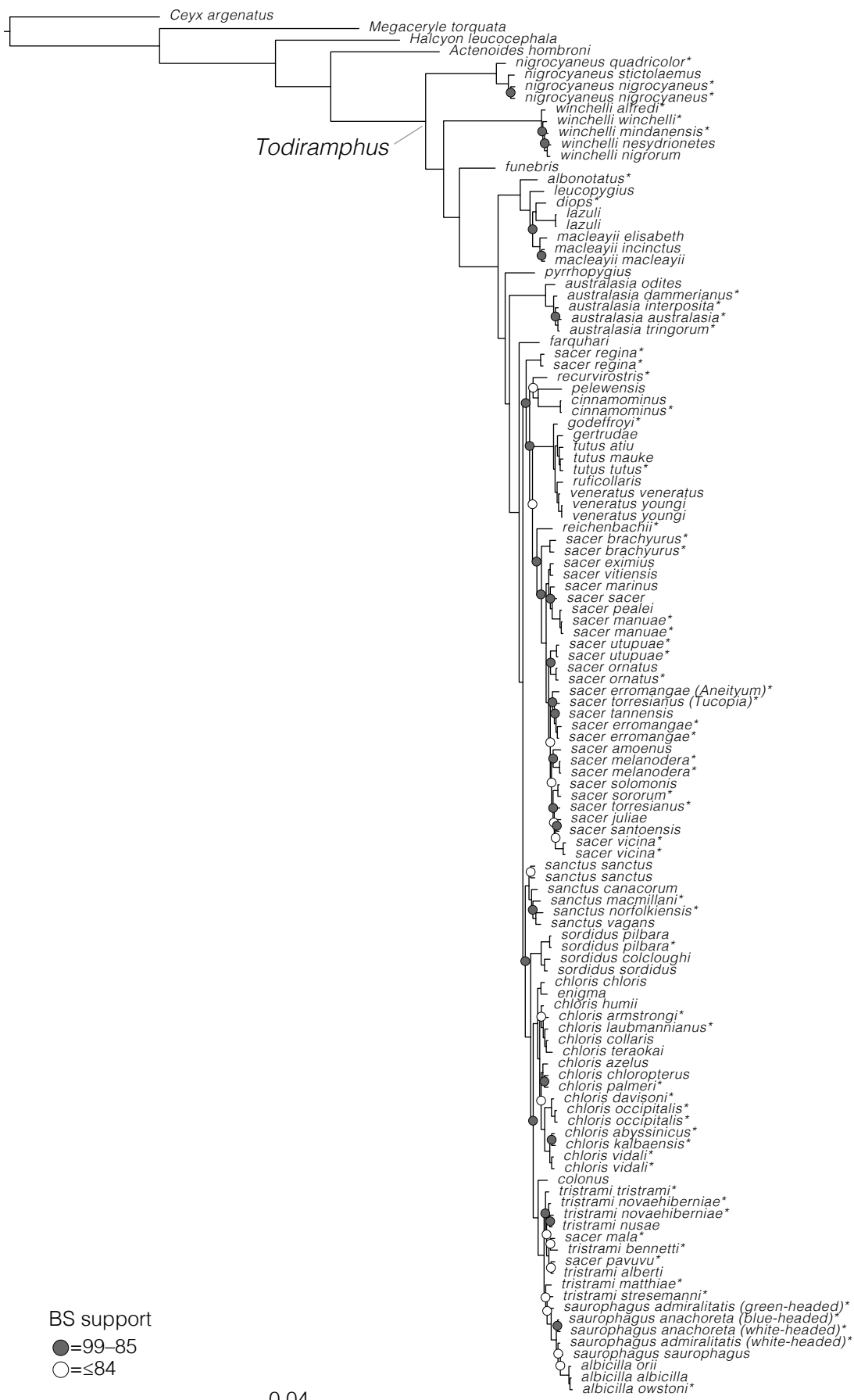


Figure S15: Maximum likelihood phylogeny, produced in IQtree, from the 100% complete SNP dataset (85,156 SNPs) for 119 tips. Circles reflect node support values that are below 100 BS: gray circles indicate 99–85 support and white circles indicate 84 or lower support. Samples with asterisks (*) indicate toepad-sourced samples.

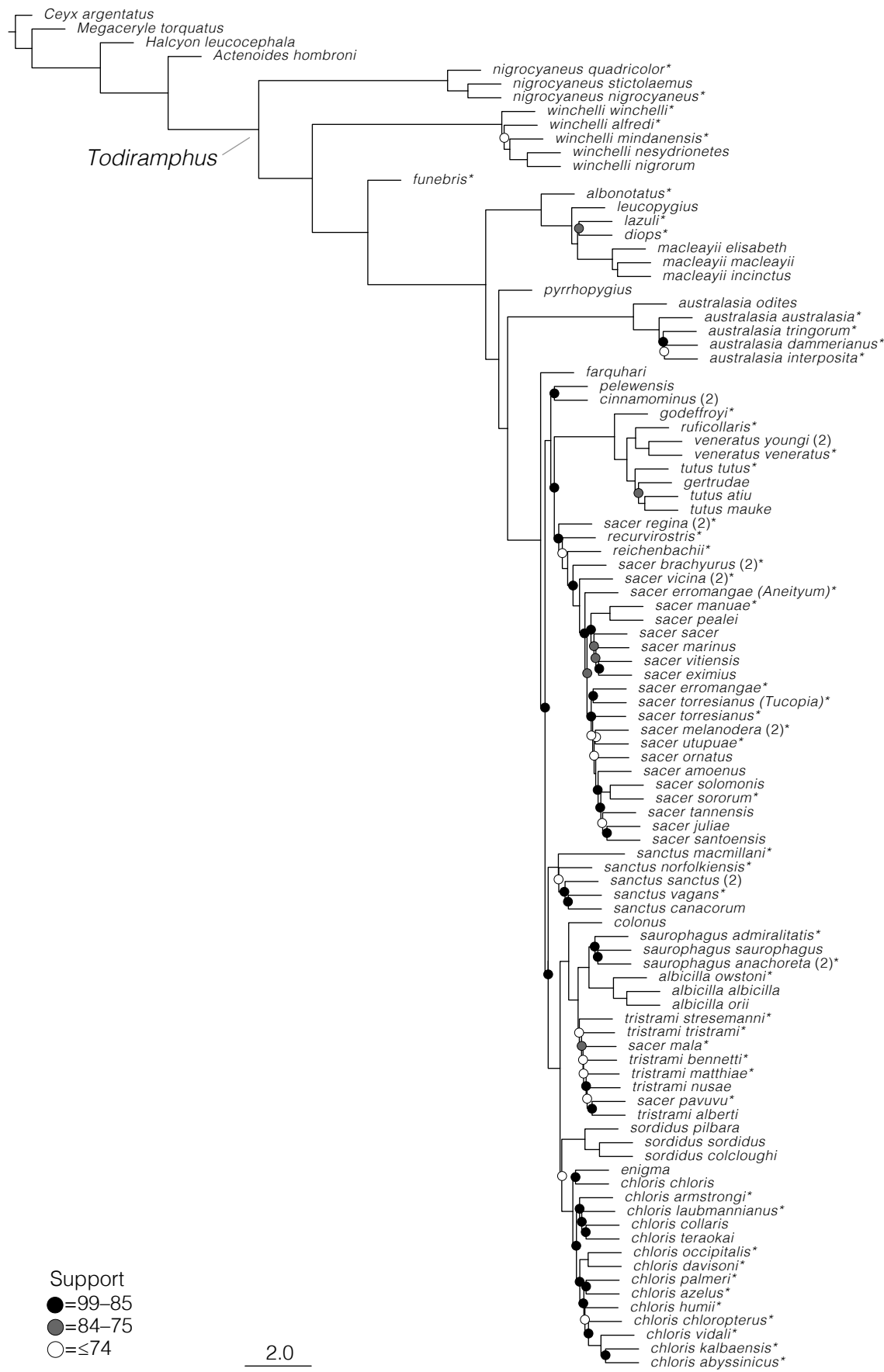


Figure S16: Species tree of *Todiramphus* kingfishers, produced in ASTRAL-III, based off of BUSCO loci that had at least 100 parsimony informative sites (470 loci, 815,282 AA sites). Circles reflect node support values that are below 100 BS: gray circles indicate 99–85 support and white circles indicate 84 or lower support. Samples with asterisks (*) indicate toepad-sourced samples.

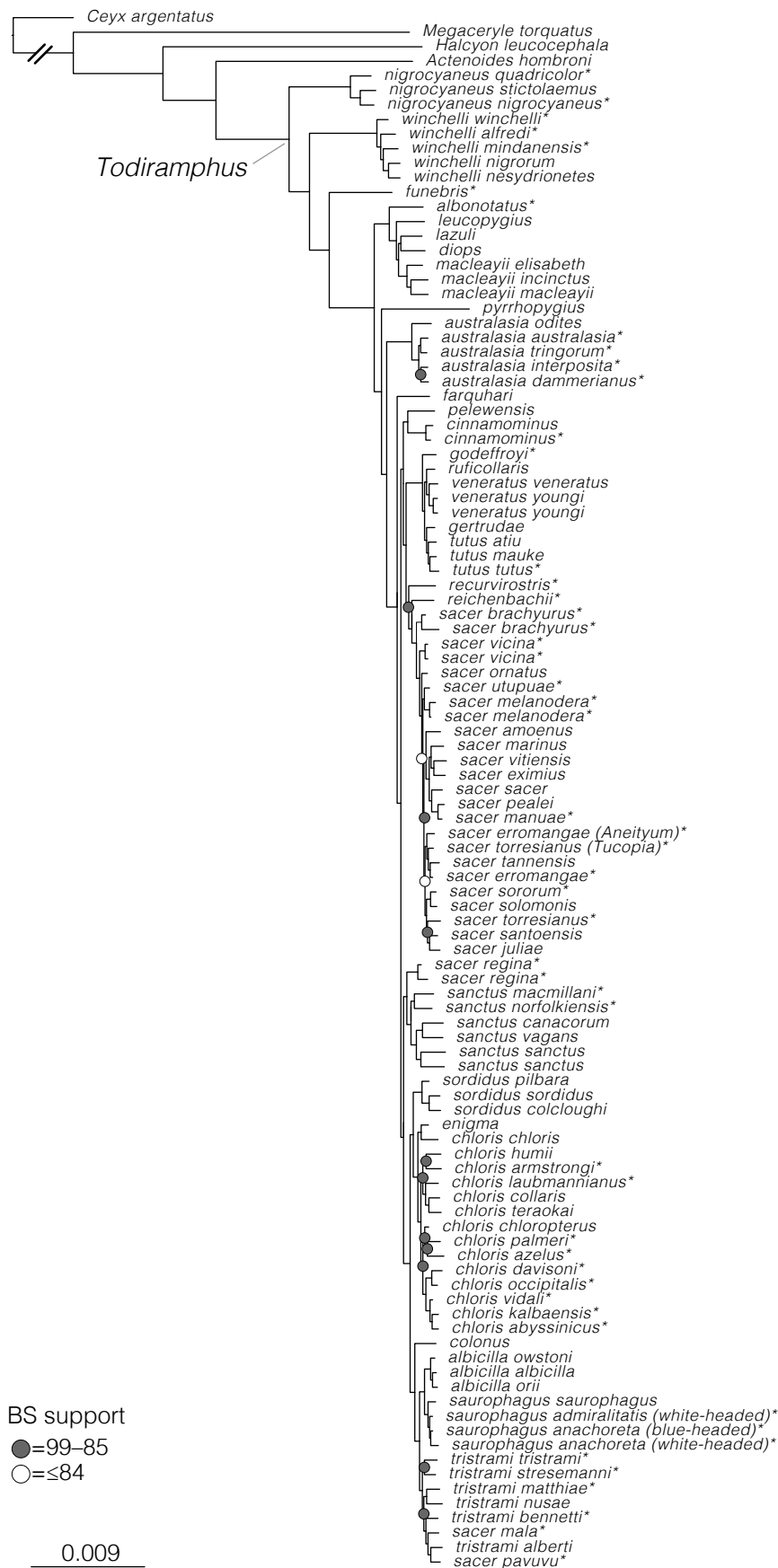


Figure S17: Partitioned maximum likelihood phylogeny, produced in IQtree, from the BUSCO dataset (8,012 loci, 5,102,241 AA sites). Circles reflect node support values that are below 100 BS: gray circles indicate 99–85 support and white circles indicate 84 or lower support. Samples with asterisks (*) indicate toepad-sourced samples.

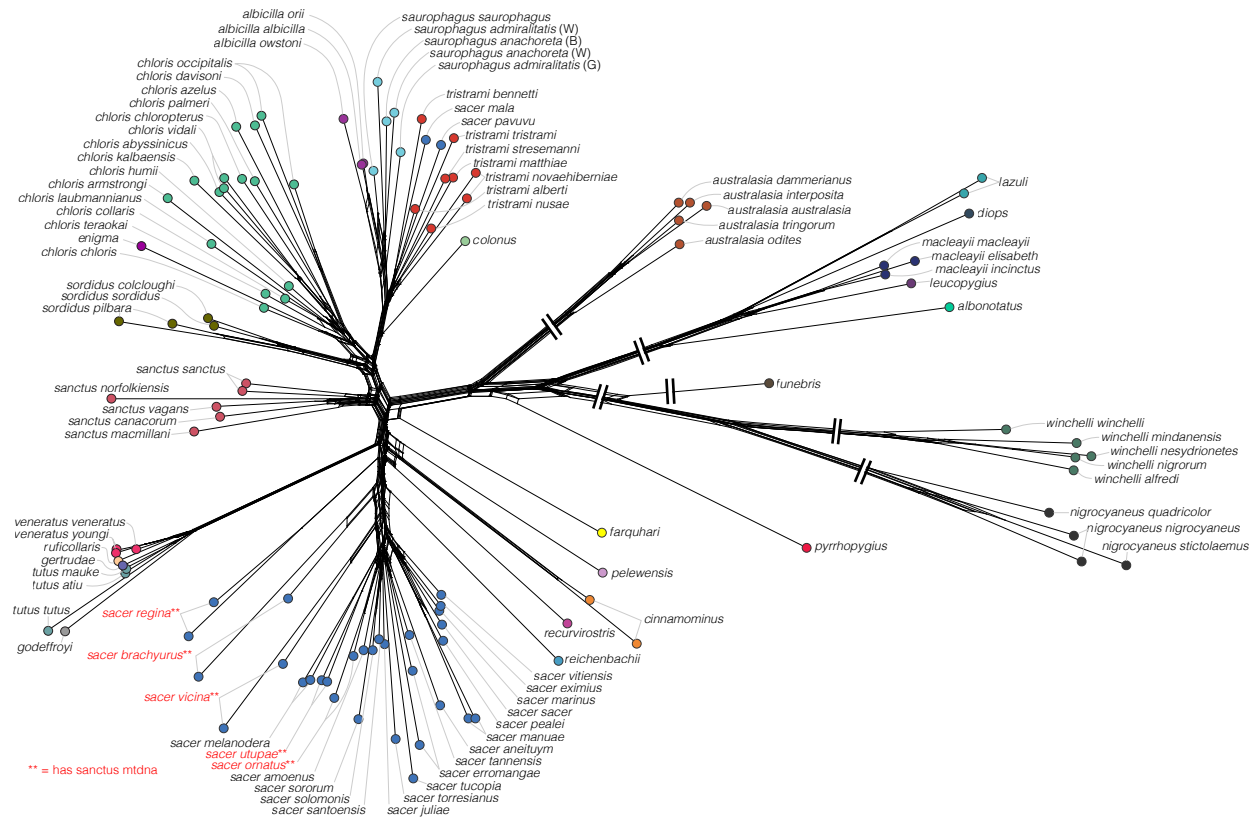


Figure S18: Unrooted network based on pairwise divergence among all samples, produced from a complete set of 85,165 SNPs. Note that some branch lengths have been artificially shortened, denoted by double slashes.

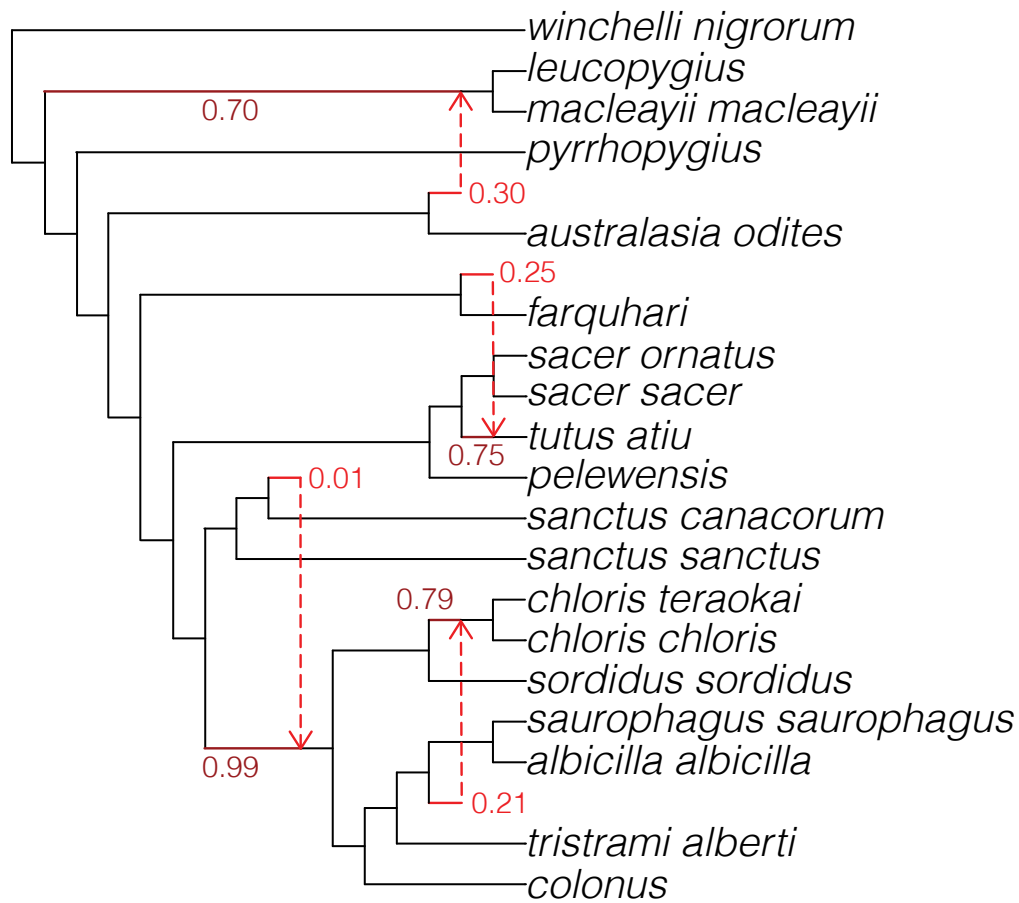
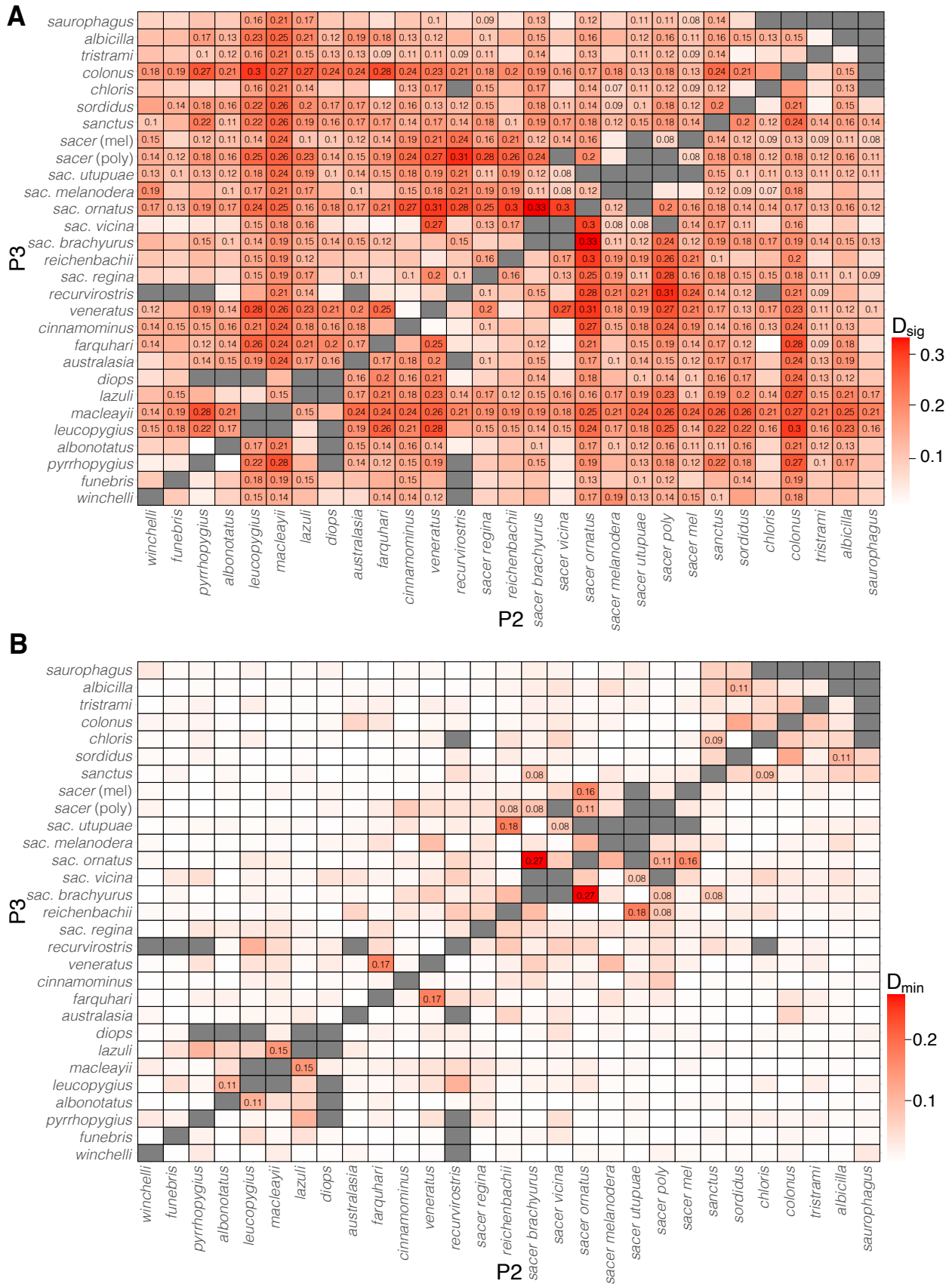


Figure S19: Phylogenetic networks estimated in SNaQ. Red arrows represent reticulation events with gamma values (the proportion of sampled genome shared via introgression). The *T. sacer ornatus* tip in this analysis is a tissue-sourced (KU 19403) sample that groups with *T. sanctus* in the mitochondrial tree (Fig. S20).



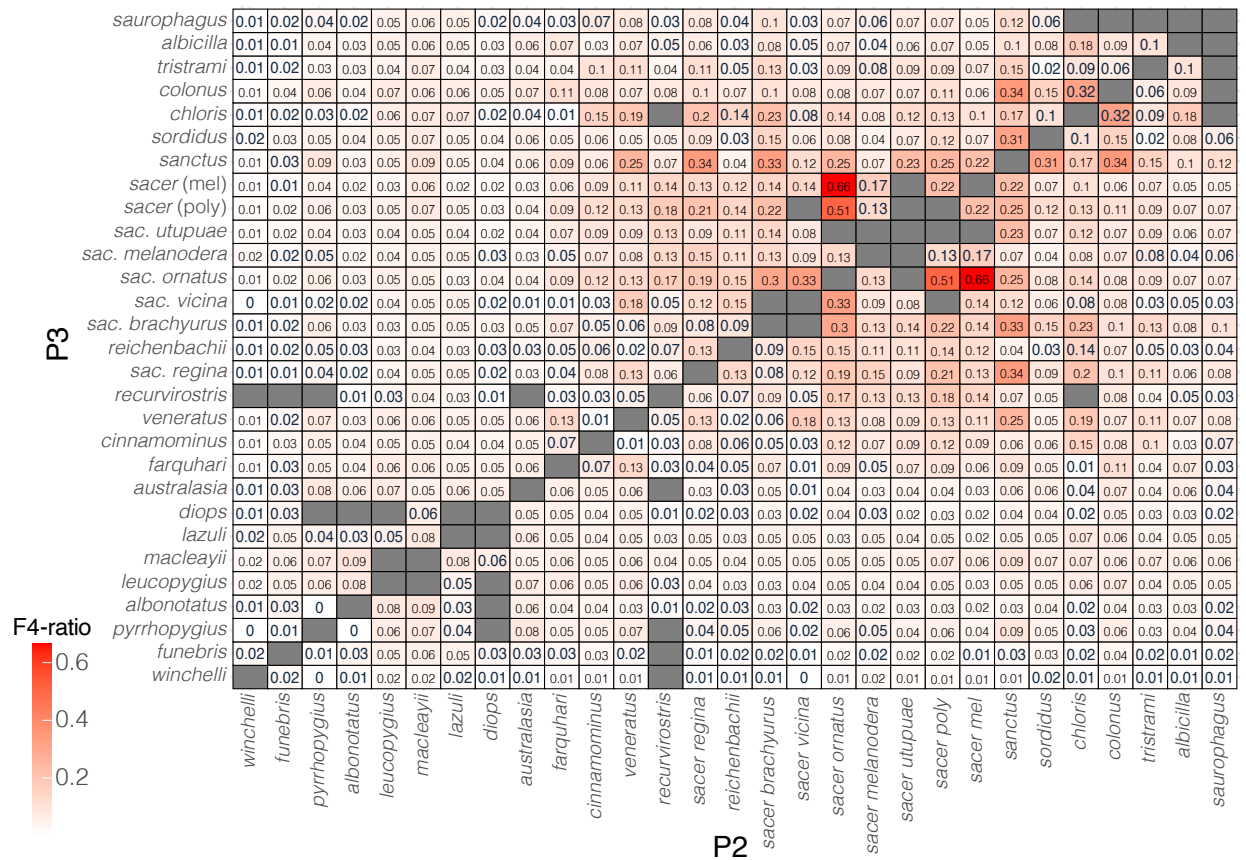


Figure S22: Pairwise results of ABBA-BABA tests, with text indicating F4-ratio values only in squares with FDR-adjusted significant pairwise p values (<0.000013). Here, *T. sacer* taxa that have *T. sanctus* mtDNA are considered their own tips and two *T. sacer* clades are those recovered in Fig. 1.

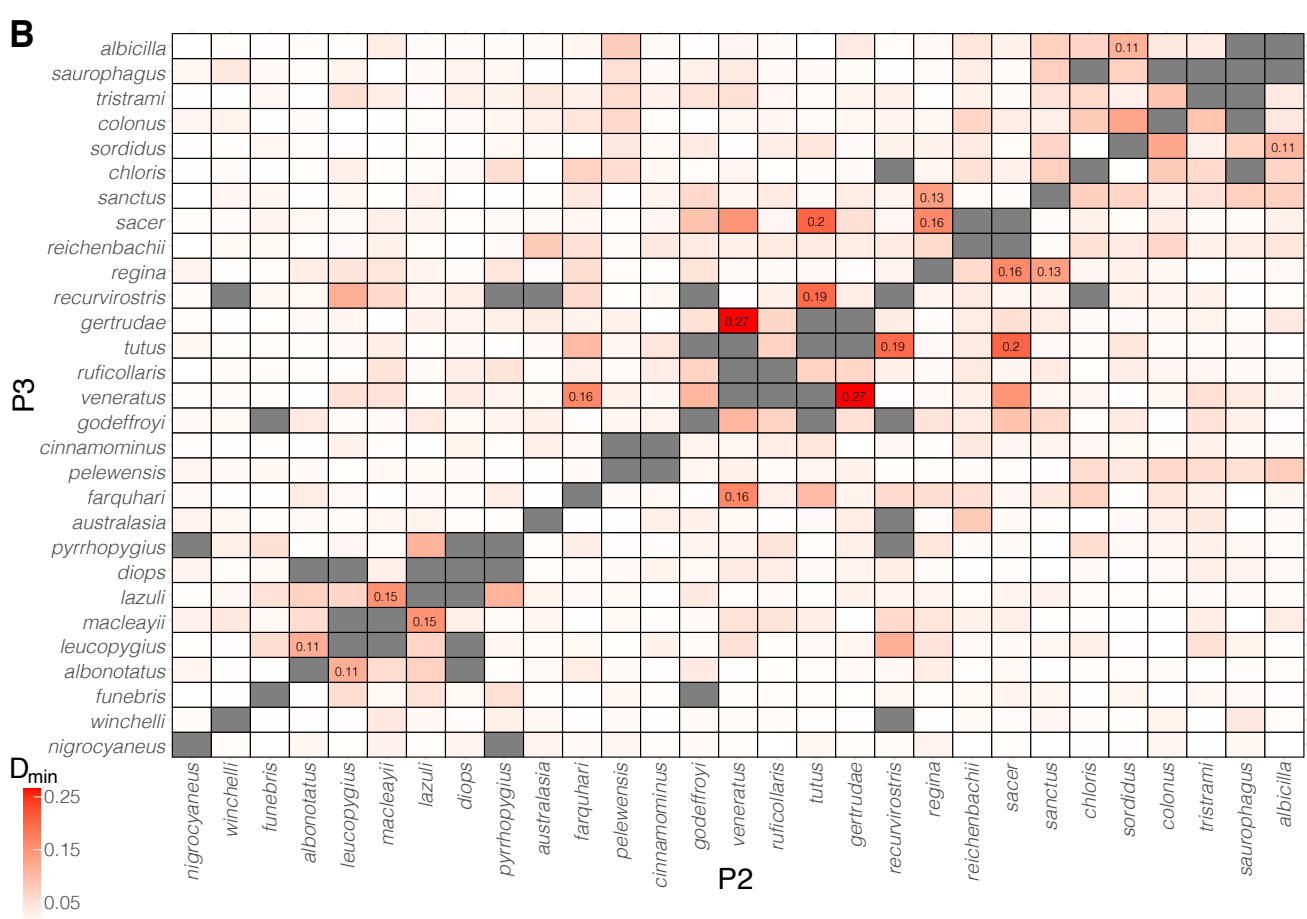
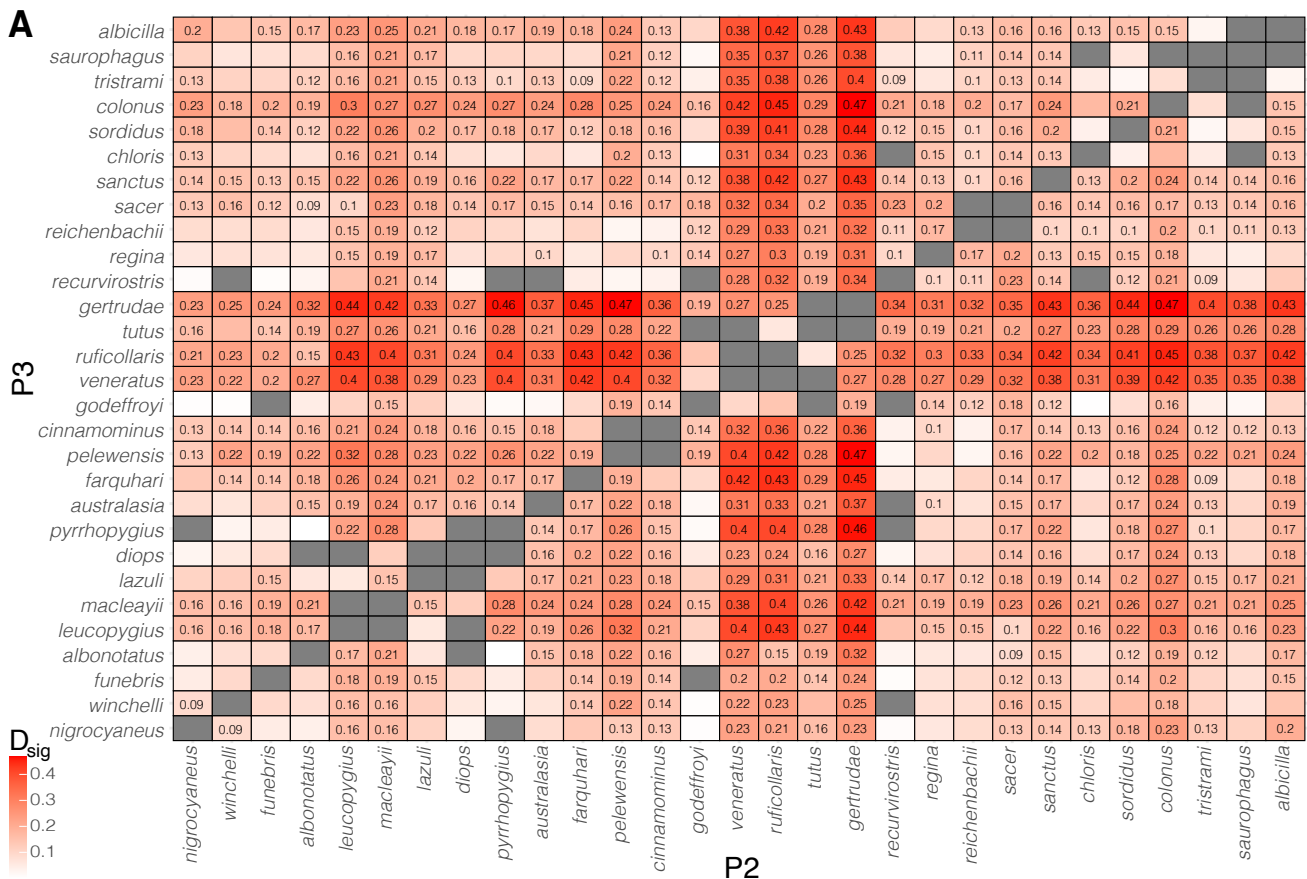


Figure S23: Pairwise results of ABBA-BABA tests, with text indicating D-statistics only in squares with FDR-adjusted significant pairwise p values (<0.000013). Panel A is the most significant (lowest pvalue) for any pairwise comparison and Panel B is the lowest value of D-statistics (Dmin). Here, only *T. s. regina* is considered its own tip and all other *T. sacer* taxa are lumped into a single tip.

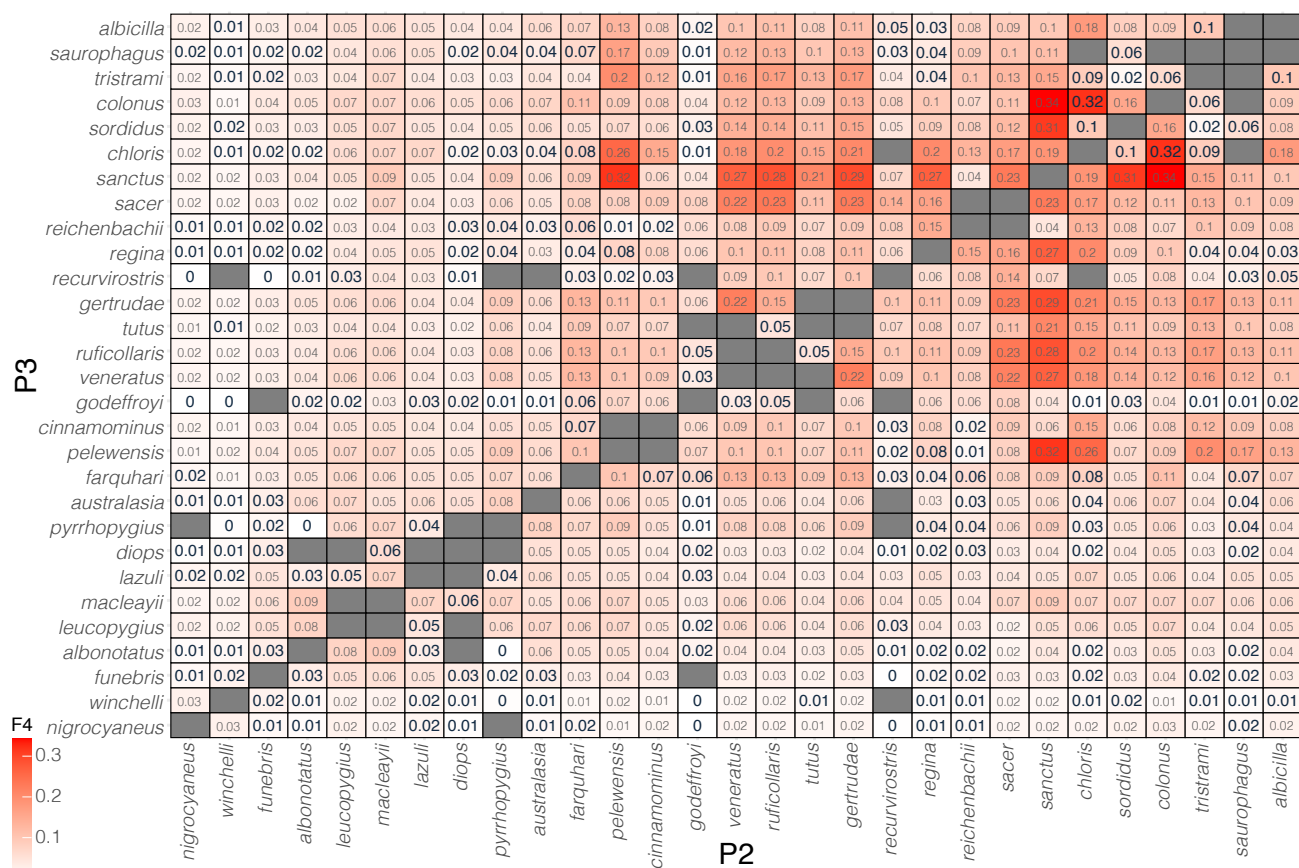


Figure S24: Pairwise results of ABBA-BABA tests, with text indicating F4-ratio values only in squares with FDR-adjusted significant pairwise p values (<0.000013). Here, only *T. s. regina* is considered its own tip and all other *T. sacer* taxa are lumped into a single tip.

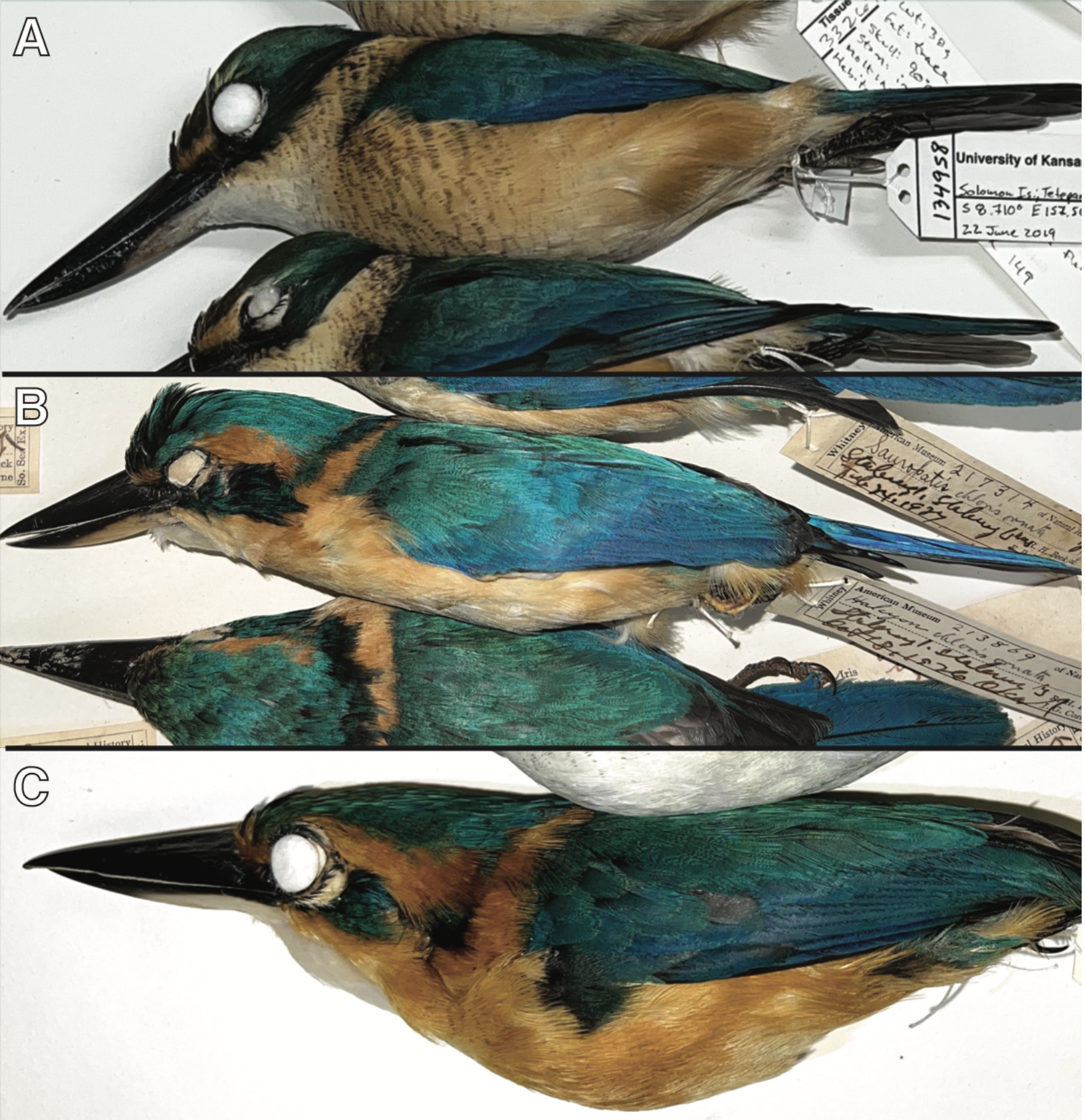


Figure S25. Comparisons of specimens of A) a representative Sacred Kingfisher (*T. sanctus sanctus*) from the University of Kansas Biodiversity Institute and Natural History Museum (KUNHM), B) a representative Pacific Kingfisher (*T. sacer ornatus*) from the American Museum of Natural History (collected during the Whitney South Seas Expedition), and C) KUNHM tissue 19403, one of the sampled *T. sacer ornatus* specimens sampled for this study. This individual grouped with other *T. sacer* in nuclear datasets but grouped with *T. sanctus* in the mitochondrial tree. Phenotypic traits that set C as a *T. sacer* is its distinctive eyestripe, and its deep buffy breast that lacks scallop patterning. Photographs by J. M. McCullough.

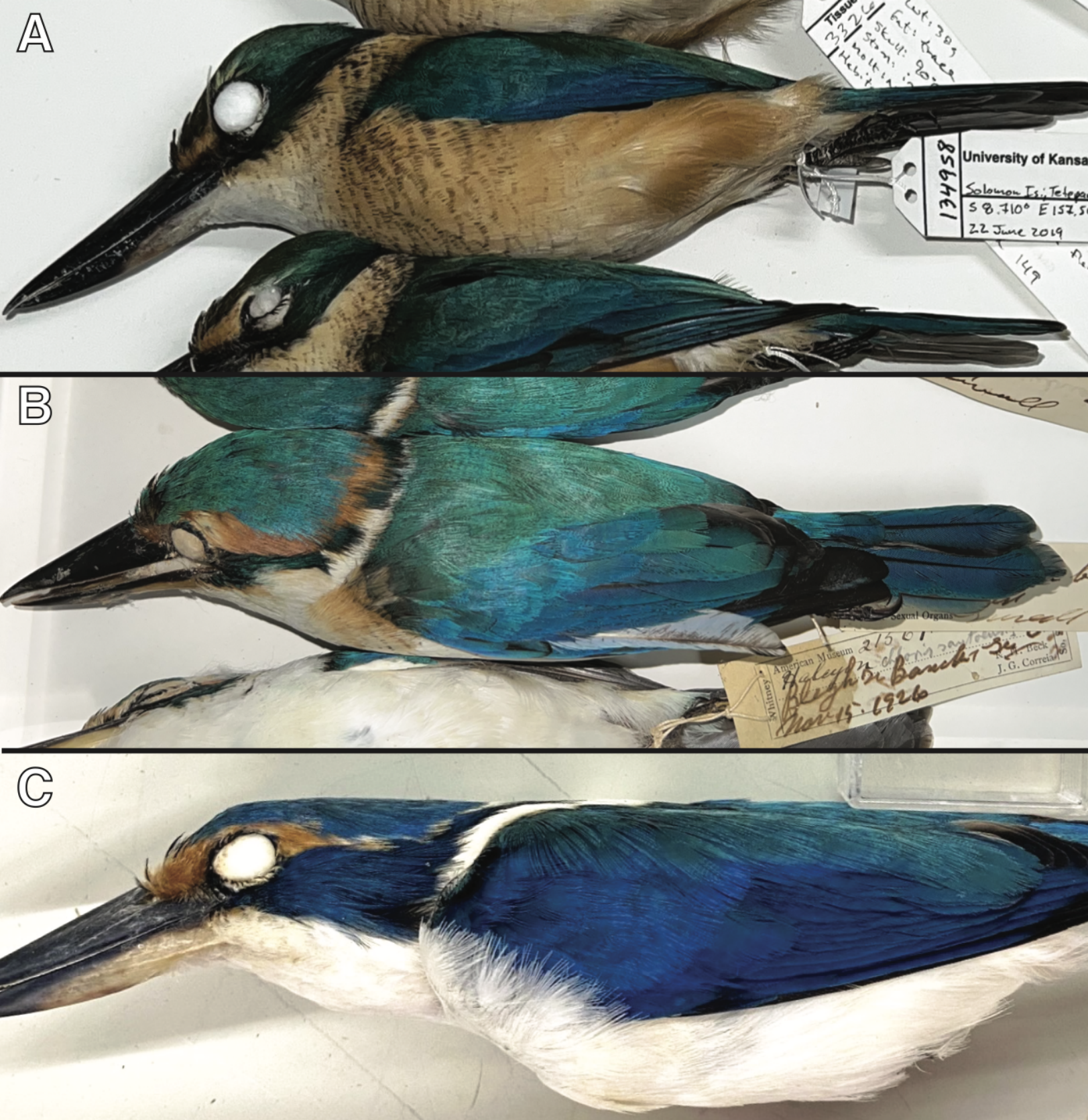


Figure S26. Comparisons of specimens of A) a representative of a Sacred Kingfisher (*T. sanctus sanctus*) from the University of Kansas Biodiversity Institute and Natural History Museum (KUNHM), B) a representative Pacific Kingfisher (*T. sacer santoensis*) from the American Museum of Natural History (collected during the Whitney South Seas Expedition), and C) LSU 45812 (skin 42929 at the University of Florida Museum of Natural History) collected from Espiritu Santo, Vanuatu. In this study, we did not sample this individual but it was mistakenly identified as a *T. sanctus* in Andersen et al. (2015) when it was inferred within the *T. sanctus* mitochondrial clade, falsely contributing to the notion that *T. sanctus* has breeding records outside of Australasia. Morphologically, it more closely resembles other *T. sacer*. Note that the blue and green tones of the backs of *Todiramphus* are often sensitive to lighting conditions and can look greener if a flash is (as in B) or not (as in C) used. Key traits that distinguish C as a *T. sacer* is its distinct eye stripe and lack of buff on its breast. Photos by J. M. McCullough (A & B) and Andrew Kratter.

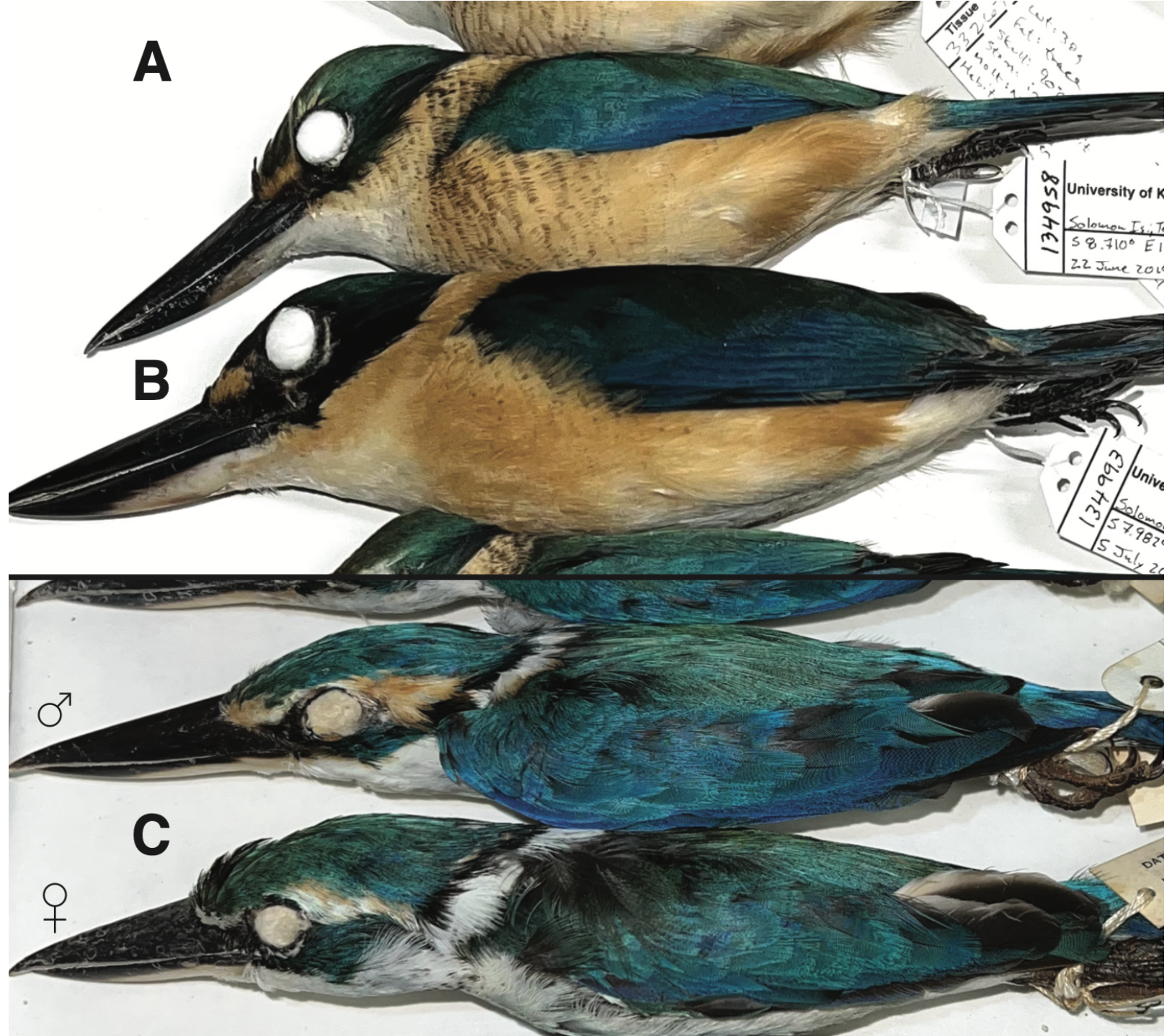


Figure S27. In its non-breeding range, *T. sanctus* can be easily misidentified as a *T. tristrami*. Comparisons of representative specimens of A) a wintering Sacred Kingfisher (*T. sanctus sanctus*) from Tetepare, Solomon Islands (collected in June 2019; KUNHM 134958), B) a resident Melanesian Kingfisher (*T. tristrami alberti*) from Kolombangara, Solomon Islands (also collected in June 2019; KUNHM 134993), and C) male and female representatives of Pacific Kingfisher (*T. sacer sororum*) from the Natural History Museum in London (collected from the Olu Malau Islands, Solomon Islands). Photographs taken by J. M. McCullough.